

First, I would like to apologize to the authors for the time it took me to complete this review.

This manuscript investigates a moderately promising approach toward improving an important aspect of reactor-monitoring safeguards using neutrinos. In particular, the potential use of inverse beta decay (IBD) directionality may have broader implications for other areas, including supernova neutrino detection, as briefly discussed by the authors. This is especially relevant given the limitations of the other existing dominant technologies in reconstructing the directionality of the neutrino component of the supernova spectrum. The authors provide a clear and appropriate contextualization of the work, explicitly state the objectives, and identify the main challenges currently facing the field. I therefore recommend the manuscript for publication in *Physical Review A*.

I have only a few minor comments, listed below.

Minor Comments

1. **Abstract:**

It would be appropriate to explicitly state that the results presented in this work are obtained for lithium-6-doped detectors. In addition, the authors should clarify that they anticipate the methodology may be generalizable to other IBD detector technologies. However, I am skeptical regarding the degree to which such generalization is valid across alternative detector implementations.

2. **Page 2:**

A more explicit discussion of the potential impact of the limited spatial distribution of lithium-6 doping in SoLid and CHANDLER detectors on the presented results would improve the clarity of the manuscript.

3. **Conclusion:**

It would be beneficial to include a more detailed discussion of the authors' intended next steps and possible future research directions arising from this work. In particular, while this paper represents a meaningful first step, it would be helpful to outline future directions aimed at improving the mathematical robustness of the method.